Causal Information Outside a Frozen Board Probe in Othello-GPT

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Abstract

Linear probes can recover Othello-GPT's board with high accuracy, and prior work has shown that changing those probe-readable features changes the model's legal-move predictions. That does not settle whether a frozen probe captures every residual-stream distinction the network uses. We study a single 8-layer Othello-GPT, a frozen 64-square linear board probe, and residual edits at `blocks.6.hook_resid_mid` that are constrained to the probe's contrast nullspace so that the probe's full occupancy-probability vector is preserved. Under that constraint, correct-target donor directions improved a legal-mass response score relative to wrong-target and random nullspace controls, both at matched residual L_2 and, in a restored protocol, at matched output Jensen–Shannon divergence of 0.01. A 64-dimensional ridge map trained only on training-split activations produced a similar target-aligned advantage without using the target donor activation at evaluation; mean absolute scores remained negative (recorded $Y_{\text{learned}} \approx -0.58$). About 56% of that map's mean improvement over a probe-space editor was already obtained by a stage-only map. On a fresh 32-triplet cohort, mean target advantages were positive, but joint switching of both arms occurred in 13 of 32 triplets, below a 60% gate. A factorial test of visible/nullspace interaction yielded estimates near zero. The results support a limited claim: this frozen probe does not exhaust the residual information that affects legal-move outputs, and some of the remainder is predictable from symbolic state, but the recovered map is not a complete transferable board. Decimal values that originate in Drive arrays were not recomputed in the local Path 1 workspace.

1 Introduction

A linear probe that reads the board from Othello-GPT is easy to treat as a description of what the model knows. Li et al. [2023] showed that nonlinear probes recover the board and that editing the representation those probes read changes legal-move predictions. Nanda et al. [2023] showed that a player-relative linear probe is enough for that reading, and that adding those probe directions steers play. Those papers give a causal role to a decoded board. They do not show that one frozen probe is a complete causal summary of the residual stream.

Probing asks whether a feature is linearly readable. An intervention asks whether a specified change to an activation changes the frozen network's output. Elazar et al. [2021] removed linearly readable information and measured behavior. The complementary experiment is to add information that a frozen probe cannot see, keep that probe's output fixed, and ask whether behavior still moves.

We run that experiment on Nanda's 8-layer Othello-GPT. We freeze a 64-square multinomial linear probe at the last real token of `blocks.6.hook_resid_mid`. Independent class-logit contrasts define a rowspace and a complementary nullspace. An activation change in the nullspace leaves every square's probe probability vector unchanged within a stated tolerance. We call that change probe-preserving relative to this probe. It does not mean that no decoder could read the change.

For recipient legal set L_r and intended-target legal set L_t, G is probability mass on squares that became legal, Q is mass on squares that became illegal, and $Y = G - Q$. A positive movement in Y is not accuracy and is not a claim that the model now plays the target board.

The contribution is empirical and narrow. Residual directions that this probe cannot distinguish still carry some target-aligned influence on legal-move predictions. A portion of that influence can be predicted from symbolic state without a donor at evaluation. The resulting intervention is not complete state transfer.

2 Related work

Readout is not use [Hewitt and Liang, 2019, Belinkov, 2022, Elazar et al., 2021]. Nullspace projection erases linearly readable features [Ravfogel et al., 2020, Belrose et al., 2023]. Othello-GPT has a causally relevant board representation [Li et al., 2023, Nanda et al., 2023]; Nanda-style steering moves *along* the probe. Activation steering adds residual directions without that kernel constraint [Turner et al., 2023, Zou et al., 2023, Rimsky et al., 2024]. Causal abstraction asks a stronger alignment question [Geiger et al., 2024, 2025] that we do not claim to answer. The measurement that remains is the kernel of one frozen board probe, with target-specific controls, with and without a donor at evaluation, including tests that fail.

3 Methods

The model is the TransformerLens LNPre conversion of 8-layer Othello-GPT ($d_{\text{model}} = 512$, vocabulary 61). We do not fine-tune it. The probe is 64 square-wise multinomial logistic regressions in a player-relative basis, trained on training activations. Independent class-logit contrasts form W. An SVD with relative tolerance 10^{-8} had recorded rank 124 and nullspace dimension 388; those integers were not recomputed from a local weight file in this package. Projectors P_R and $P_N = I - P_R$ satisfy $WP_N = 0$.

Softmax is over 61 tokens. JS uses natural-log KL after adding 10^{-12}. SPEC froze 48 pairs (hash `2789accf…`). NST froze a different 48 unused-validation pairs (hash `39abdc79…`). FRESH used 32 triplets (hash `3ae44e1e…`). VNF used a new 32-triplet cohort (hash `d7ceb056…`). Confirmation games were not opened.

Donor-null interventions use $\delta = P_N(h_D - h_E)$. Equal-JS search calibrates each direction independently to JS $= 0.01 \pm 0.0002$. The original COST run violated that protocol; only the restoration is treated as equal-JS. The learned map predicts $k = 64$ PCA coefficients of $P_N h$ from symbolic features, including the true legal mask, with ridge $\alpha = 10$ selected on training-game CV R^2, never held-out Y. Evaluation uses $\widehat{z}(b_t) - P_N h_E$. Bootstrap intervals are clustered on recipient game ($n = 48$) or triplet ($n = 32$).

4 Results

FIBER supported probe-logit insufficiency: nullspace edits can change outputs while probe probabilities stay within tolerance. SPEC, at matched $L_2 \approx 18.33$, recorded $\Delta_{\text{specific}} = +0.391627$. Random directions of that length barely moved JS (≈ 0.038 versus 0.279 for correct donors), so unmatched disruption is a confound.

The original COST equal-JS attempt remains `TARGET_SPECIFIC_AT_EQUAL_JS_INCONCLUSIVE`. The restored independent calibration on the same 48 pairs recorded $\Delta_{\text{wrong}} = +0.02655$ and

$\Delta_{\text{random}} = +0.02297$ at JS = 0.01, with unequal residual L_2 (5.30 / 6.37 / 12.27). Letter: `FIXED_JS_TARGET_SPECIFICITY_SUPPORTED`. This is not sealed confirmation.

NST recorded $Y_{\text{learned}} - Y_{\text{wrong}} = 0.020057$ $[0.010791, 0.029784]$ with $Y_{\text{learned}} = -0.582$ still negative. Letter: `DONOR_FREE_NULL_STATE_TRANSFER_SUPPORTED`. Donor-free is an evaluation-time statement. A gradient baseline calibrated on 7/48 pairs is not a full-sample comparator.

Exploratory ablations found FULL−LEGAL_ONLY and FULL−BOARD_ONLY below a +0.01 floor (`LEGAL_MASK_SUFFICES_IN_TESTED_FAMILY`). A git-tracked mean identity shows STAGE_ONLY already provides $0.030099/0.053713 \approx 56\%$ of FULL's mean improvement over Editor A. That is a ratio of means, not an independent causal decomposition.

FRESH natural-anchor FULL had $A_T = +0.02850$ and $A_W = +0.01856$, but both-positive switching in 13/32 triplets ($40.6\% < 60\%$). Letter: `SYMMETRIC_TARGET_SWITCHING_NOT_SUPPORTED`. VNF recorded $I_T = 0.000808$ and $I_W = 0.000972$, both missing +0.005, intervals including zero. Letter: `TARGET_SPECIFIC_VISIBLE_NULL_INTERACTION_NOT_SUPPORTED`. 08R2 Editor A validation remains `NEGATIVE`.

5 Discussion and limitations

This frozen probe's rowspace is not the whole residual story for legal-move mass. The kernel effect is small after JS matching, partly stage-conditioned, and not a transferable board. Off-manifold risk is untested. The legal mask is supplied, not inferred. SPEC and fixed-JS share pairs. Sealed confirmation was not used. Arrays were not recomputed in the local package workspace. Nothing here is evidence about large language models.

6 Conclusion

A frozen linear board probe on Othello-GPT can hold its full occupancy-probability vector fixed while residual edits in its contrast nullspace still change legal-move predictions in a target-aligned way, including without a donor at evaluation. Absolute scores stay negative, joint switching fails, and a factorial visible/null interaction is unsupported. This probe is not a sufficient causal description of the residual information that affects legal-move outputs. It is not a second world model.

Reproducibility

See `REPRODUCIBILITY.md` in the Path 1 directory. Official letters listed in the markdown manuscript are preserved. AI assistance was used in drafting; an AI system is not an author. Human authors remain unassigned.

References

Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.

Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems*, 2023. URL `https://proceedings.neurips.cc/paper_files/paper/2023/hash/d066d21c619d0a78c5b557fa3291a8f4-Abstract-Conference.html`.

Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021. doi: 10.1162/tacl_a_00359. URL https://aclanthology.org/2021.tacl-1.10/.

Atticus Geiger, Zhengxuan Wu, Christopher Potts, Thomas Icard, and Noah Goodman. Finding alignments between interpretable causal variables and distributed neural representations. In *Proceedings of the Third Conference on Causal Learning and Reasoning*, volume 236 of *Proceedings of Machine Learning Research*, pages 160–187, 2024. URL https://proceedings.mlr.press/v236/geiger24a.html.

Atticus Geiger, Duligur Ibeling, Amir Zur, Maheep Chaudhary, Sonakshi Chauhan, Jing Huang, Aryaman Arora, Zhengxuan Wu, Noah Goodman, Christopher Potts, and Thomas Icard. Causal abstraction: A theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research*, 26(83):1–64, 2025. URL https://jmlr.org/papers/v26/23-0058.html.

John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 2019. doi: 10.18653/v1/D19-1275.

Kenneth Li, Aspen K. Hopkins, David Bau, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Emergent world representations: Exploring a sequence model trained on a synthetic task. In *International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2210.13382.

Neel Nanda, Andrew Lee, and Martin Wattenberg. Emergent linear representations in world models of self-supervised sequence models. In *Proceedings of the 6th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 16–30. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.blackboxnlp-1.2. URL https://aclanthology.org/2023.blackboxnlp-1.2/.

Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7237–7256, 2020. doi: 10.18653/v1/2020.acl-main.647. URL https://aclanthology.org/2020.acl-main.647/.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Turner. Steering llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522, 2024. doi: 10.18653/v1/2024.acl-long.828. URL https://aclanthology.org/2024.acl-long.828/.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023. URL https://arxiv.org/abs/2308.10248.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023. URL https://arxiv.org/abs/2310.01405. Author list copied from arXiv 2310.01405; recheck the current arXiv version before any public release.